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ASTRO ALLIANCE

DATA DICTIONARY AND FEATURE DOCUMENTATION

Comprehensive Guide to the Variables in the Exoplanet Dataset

1. **pl_rade (Planet Radius [Earth Radii] Target Variable 1)** = The radius of the exoplanet expressed as a multiple of Earth's radius.
2. **pl_bmasse (Planet Best Mass [Earth Masses] Target Variable 2)** = The best estimate of the mass of the planet, measured in units of Earth masses. This is typically the measured value (M) or minimum mass ($M \sin i$) depending on the discovery technique.
3. **pl_orbper (Orbital Period [Days])** = The amount of time it takes for the exoplanet to complete one full orbit around its host star, measured in solar days.
4. **pl_orbpererr1 (Orbital Period Upper Uncertainty)** = The positive (upper) margin of error or measurement uncertainty for the planet's orbital period.
5. **pl_orbpererr2 (Orbital Period Lower Uncertainty)** = The negative (lower) margin of error or measurement uncertainty for the planet's orbital period.
6. **pl_orbsmax (Orbit Semi-Major Axis [AU])** = The semi-major axis of the planet's elliptical orbit, which represents the average distance from the host star in Astronomical Units (AU).
7. **pl_orbsmaxerr1 (Orbit Semi-Major Axis Upper Uncertainty)** = The positive margin of error for the planet's semi-major axis measurement.
8. **pl_orbsmaxerr2 (Orbit Semi-Major Axis Lower Uncertainty)** = The negative margin of error for the planet's semi-major axis measurement.
9. **pl_orbeccen (Eccentricity)** = The amount by which the planet's orbit deviates from a perfect circle (0 = perfect circle, values closer to 1 signify highly elongated orbits).
10. **pl_orbincl (Orbital Inclination [Deg])** = The angle of the planet's orbital plane relative to the plane of the sky or the reference line from Earth, measured in degrees.
11. **pl_insol (Insolation Flux [Earth Flux])** = The amount of stellar energy received by the planet per unit area compared to the energy Earth receives from the Sun.
12. **pl_eqt (Equilibrium Temperature [K])** = The theoretical temperature calculated for the planet assuming it acts as a blackbody heated solely by its host star, measured in Kelvin.



13. **pl_imppar (Impact Parameter)** = The minimum distance between the center of the stellar disk and the center of the planet's trajectory during a transit event, normalized by stellar radius.
14. **pl_trandur (Transit Duration [Hours])** = The duration of time it takes for a transiting planet to pass completely across the face of its host star, measured in hours.
15. **pl_ratror (Ratio of Planet Radius to Stellar Radius)** = The dimensionless size ratio of the planet's radius divided by its host star's radius. Essential for transit depth analysis.
16. **pl_ratrorerr1 (Ratio of Planet Radius to Stellar Radius Upper Uncertainty)** = The positive margin of error for the radius ratio.
17. **pl_ratrorerr2 (Ratio of Planet Radius to Stellar Radius Lower Uncertainty)** = The negative margin of error for the radius ratio.
18. **st_mass (Stellar Mass [Solar Masses])** = The mass of the host star expressed as a multiple of our Sun's mass.
19. **st_masserr1 (Stellar Mass Upper Uncertainty)** = The positive margin of error for the host star's mass.
20. **st_masserr2 (Stellar Mass Lower Uncertainty)** = The negative margin of error for the host star's mass.
21. **st_rad (Stellar Radius [Solar Radii])** = The radius of the host star expressed as a multiple of our Sun's radius.
22. **st_raderr1 (Stellar Radius Upper Uncertainty)** = The positive margin of error for the host star's radius.
23. **st_raderr2 (Stellar Radius Lower Uncertainty)** = The negative margin of error for the host star's radius.
24. **st_teff (Stellar Effective Temperature [K])** = The surface temperature of the host star determined from its spectral characteristics, measured in Kelvin.
25. **st_tefferr1 (Stellar Effective Temperature Upper Uncertainty)** = The positive margin of error for the stellar surface temperature.
26. **st_tefferr2 (Stellar Effective Temperature Lower Uncertainty)** = The negative margin of error for the stellar surface temperature.
27. **st_logg (Stellar Surface Gravity [$\log_{10}(\text{cm/s}^2)$])** = The base-10 logarithm of the gravitational acceleration at the surface of the host star.
28. **st_loggerr1 (Stellar Surface Gravity Upper Uncertainty)** = The positive margin of error for the stellar surface gravity.
29. **st_loggerr2 (Stellar Surface Gravity Lower Uncertainty)** = The negative margin of error for the stellar surface gravity.



30. **st_met (Stellar Metallicity [dex])** = The abundance of elements heavier than hydrogen and helium present in the star, relative to the Sun's metallicity (expressed on a logarithmic scale).
31. **st_meterr1 (Stellar Metallicity Upper Uncertainty)** = The positive margin of error for the stellar metallicity.
32. **st_meterr2 (Stellar Metallicity Lower Uncertainty)** = The negative margin of error for the stellar metallicity.
33. **st_dens (Stellar Density [g/cm³])** = The average density of the host star, calculated in grams per cubic centimeter.
34. **sy_vmag (System V-band Magnitude)** = The apparent brightness of the planetary system measured through a standard visual (V) optical filter. Lower numbers mean brighter objects.
35. **sy_vmagerr1 (System V-band Magnitude Upper Uncertainty)** = The positive margin of error for the system's V-band magnitude.
36. **sy_vmagerr2 (System V-band Magnitude Lower Uncertainty)** = The negative margin of error for the system's V-band magnitude.
37. **sy_bmag (System B-band Magnitude)** = The apparent brightness of the planetary system in the blue (B) optical filter spectrum.
38. **sy_bmagerr1 (System B-band Magnitude Upper Uncertainty)** = The positive margin of error for the system's B-band magnitude.
39. **sy_bmagerr2 (System B-band Magnitude Lower Uncertainty)** = The negative margin of error for the system's B-band magnitude.
40. **sy_jmag (System J-band Magnitude)** = The apparent brightness of the planetary system measured in the near-infrared J-band wavelength (~1.25 micrometers).
41. **sy_jmagerr1 (System J-band Magnitude Upper Uncertainty)** = The positive margin of error for the system's J-band magnitude.
42. **sy_jmagerr2 (System J-band Magnitude Lower Uncertainty)** = The negative margin of error for the system's J-band magnitude.
43. **sy_hmag (System H-band Magnitude)** = The apparent brightness of the planetary system in the near-infrared H-band wavelength (~1.65 micrometers).
44. **sy_hmagerr1 (System H-band Magnitude Upper Uncertainty)** = The positive margin of error for the system's H-band magnitude.
45. **sy_hmagerr2 (System H-band Magnitude Lower Uncertainty)** = The negative margin of error for the system's H-band magnitude.
46. **sy_kmag (System K-band Magnitude)** = The apparent brightness of the planetary system in the near-infrared K-band wavelength (~2.2 micrometers).



47. **sy_kmagerr1 (System K-band Magnitude Upper Uncertainty)** = The positive margin of error for the system's K-band magnitude.
48. **sy_kmagerr2 (System K-band Magnitude Lower Uncertainty)** = The negative margin of error for the system's K-band magnitude.
49. **sy_w1mag (System WISE W1 Magnitude)** = The apparent brightness of the planetary system in the mid-infrared W1 band (~3.4 micrometers) measured by the Wide-field Infrared Survey Explorer.
50. **sy_w1magerr1 (System WISE W1 Magnitude Upper Uncertainty)** = The positive margin of error for the W1 magnitude.
51. **sy_w1magerr2 (System WISE W1 Magnitude Lower Uncertainty)** = The negative margin of error for the W1 magnitude.
52. **sy_w2mag (System WISE W2 Magnitude)** = The apparent brightness of the planetary system in the mid-infrared W2 band (~4.6 micrometers) from the WISE telescope.
53. **sy_w2magerr1 (System WISE W2 Magnitude Upper Uncertainty)** = The positive margin of error for the W2 magnitude.
54. **sy_w2magerr2 (System WISE W2 Magnitude Lower Uncertainty)** = The negative margin of error for the W2 magnitude.
55. **sy_w3mag (System WISE W3 Magnitude)** = The apparent brightness of the planetary system in the thermal-infrared W3 band (~12 micrometers) from the WISE telescope.
56. **sy_w3magerr1 (System WISE W3 Magnitude Upper Uncertainty)** = The positive margin of error for the W3 magnitude.
57. **sy_w3magerr2 (System WISE W3 Magnitude Lower Uncertainty)** = The negative margin of error for the W3 magnitude.
58. **sy_w4mag (System WISE W4 Magnitude)** = The apparent brightness of the planetary system in the far-infrared W4 band (~22 micrometers) from the WISE telescope.
59. **sy_gaiamag (System Gaia Magnitude)** = The apparent brightness of the system as measured across the broad optical band by the European Space Agency's Gaia spacecraft.
60. **sy_tmag (System TESS Magnitude)** = The apparent brightness of the system in the specialized broad red/near-infrared band used by NASA's Transiting Exoplanet Survey Satellite.
61. **sy_snum (Number of Stars in System)** = The total number of cataloged stars belonging to this particular planetary system (e.g., binary, tertiary, or single star configurations).
62. **sy_pnum (Number of Planets in System)** = The total number of confirmed exoplanets discovered orbiting within this planetary system.
63. **sy_mnum (Number of Moons in System)** = The total number of confirmed or candidate natural satellites (exomoons) discovered around the exoplanets in this system.

64. **discoverymethod (Discovery Method)** = The astronomical technique used to originally detect and confirm the exoplanet (e.g., Transit, Radial Velocity, Direct Imaging, Microlensing).
65. **disc_year (Discovery Year)** = The calendar year in which the exoplanet's discovery was officially confirmed and published in scientific literature.
66. **disc_facility (Discovery Facility)** = The specific observatory, ground station, or space-based telescope mission responsible for detecting the exoplanet (e.g., Kepler, TESS, W. M. Keck Observatory).
67. **soltype (Solution Type)** = The cataloging tag designating the nature of the data entry solution or the processing pipeline settings applied to the planetary record.
68. **sy_dist (System Distance [Parsecs])** = The distance from Earth to the planetary system, measured in units of parsecs (1 parsec $\approx$ 3.26 light-years).
69. **sy_disterr1 (System Distance Upper Uncertainty)** = The positive margin of error for the distance parameter.
70. **sy_disterr2 (System Distance Lower Uncertainty)** = The negative margin of error for the distance parameter.
71. **sy_plx (System Parallax [mas])** = The stellar parallax angle measured in milliarcseconds, which forms the trigonometric basis for calculating interstellar distance from Earth.
72. **sy_pm (System Total Proper Motion [mas/yr])** = The overall angular velocity of the system across the sky relative to distant background stars, measured in milliarcseconds per year.
73. **sy_pmra (System Proper Motion (RA) [mas/yr])** = The component of the system's proper motion along the direction of Right Ascension.
74. **sy_pmdec (System Proper Motion (Dec) [mas/yr])** = The component of the system's proper motion along the direction of Declination.
75. **glon (Galactic Longitude [Deg])** = The angular coordinate of the planetary system measured eastward along the galactic equator from the Galactic Center, in degrees.
76. **glat (Galactic Latitude [Deg])** = The angular distance of the planetary system north or south of the plane of the Milky Way galaxy, in degrees.
77. **cosmic_noise_1 (Cosmic Noise Filter Feature 1)** = A synthesized or simulated noise threshold used in the analytics pipeline to account for stellar activity or instrument artifacts.
78. **cosmic_noise_2 (Cosmic Noise Filter Feature 2)** = A secondary diagnostic noise variable used for regularizing or validating model features.
79. **pl_orbper_noisy (Noisy Orbital Period)** = A perturbed or synthetically modified version of the orbital period feature, typically utilized for training robustness testing or data augmentations.
80. **pl_orbsmax_noisy (Noisy Semi-Major Axis)** = A perturbed or artificially altered version of the semi-major axis feature, used to benchmark analytical models against feature variation.



81. **orbital_size_score (Orbital Size Score)** = A calculated engineered score reflecting the geometric scale of the orbit relative to standard reference systems.
82. **stellar_scale_score (Stellar Scale Score)** = An engineered metric summarizing the dimensions and mass traits of the host star for simplified multi-variable grouping.
83. **rowid (Row Identifier)** = The unique internal integer index mapping each row record in this data matrix.
84. **pl_name (Planet Name)** = The official systematic designation given to the exoplanet (usually consisting of the host star name followed by a lowercase letter starting with 'b').
85. **hostname (Host Star Name)** = The primary scientific name of the parent star or multiple-star system around which the exoplanet orbits.
86. **pl_letter (Planet Letter Designation)** = The lowercase letter assigned to the exoplanet in order of discovery within its system (e.g., b, c, d).
87. **rowupdate (Row Last Update Timestamp)** = The calendar date when this planetary entry was last updated or verified in the data archive database.
88. **pl_pubdate (Planet Parameters Publication Date)** = The year and month corresponding to the primary scientific paper publication from which these planet properties were extracted.
89. **releasedate (Data Release Date)** = The administrative date when this specific entry row was made publicly available in the integrated analytics dataset.
90. **disc_pubdate (Discovery Publication Date)** = The scientific publication date tracking the original research article that declared the formal confirmation of the planet.
91. **st_nphot (Number of Photometric Observations)** = The total count of distinct light curve or photometric measurement sequences available for the host star in public.
92. **st_nrvc (Number of Radial Velocity Observations)** = The total number of high-precision radial velocity or Doppler shift spectra cataloged for this host star.
93. **st_nspec (Number of Stellar Spectra Books)** = The total number of atmospheric or chemical spectral profiles captured and archived for the parent star.
94. **pl_nespec (Number of Emission Spectra Datasets)** = The count of atmospheric emission spectra datasets collected for the planet during secondary eclipses or phase changes.
95. **pl_ntranspec (Number of Transmission Spectra Datasets)** = The count of atmospheric transmission spectra records obtained during transits to probe the chemical composition of the exoplanet's limbs.
96. **pl_ndispec (Number of Direct Imaging Spectra)** = The total number of direct imaging or coronagraphic spectroscopic files resolved for this exoplanet.